

Beyond Humanity’s Last Exam: A Survey on LLM-Driven Difficult Q&A Synthesis

Abstract

The rapid advancement of large language models (LLMs) has rendered static benchmarks increasingly insufficient for evaluating frontier AI capabilities. Consequently, the automated and semi-automated synthesis of high-difficulty, contamination-resistant assessment items—particularly those approaching the rigor of “Humanity’s Last Exam” (HLE)—has emerged as a critical research frontier. This survey systematically reviews the state of the art, covering three interconnected pillars: (i) automated generation pipelines, from prompt engineering and chain-of-thought manipulation to concrete implemented systems including hypothesis-driven error analysis, self-play evaluation, multi-agent verification, and automated benchmark auditing; (ii) human-in-the-loop (HITL) paradigms that integrate expert judgment through RLHF variants, explicit collaborative frameworks, and multi-agent verification architectures; and (iii) domain-specific applications in mathematics, physics, chemistry, and materials science, including tool-augmented AI scientist agents. We further examine the co-evolving evaluation ecosystem, with particular attention to empirical contamination analysis and the meta-evaluation problem. Our analysis reveals a paradigm shift from one-shot prompt-based generation toward complex, multi-stage, multi-agent “creation ecosystems” that synergistically combine LLM scalability with human cognitive depth, building on foundational insights from the broader synthetic data generation literature.

Keywords: Large Language Models; Difficult Question Synthesis; Human-in-the-Loop; Benchmark Generation; Reinforcement Learning from Human Feedback; Curriculum Learning; AI Evaluation

Contents

1	Introduction	3
1.1	Background and Motivation	3
1.2	Scope and Research Questions	3
1.3	Relation to Existing Surveys	4
1.4	Contributions and Organization	4
2	Theoretical Foundations and Core Challenges	4
2.1	Defining “Difficulty”: A Multi-Dimensional Framework	4
2.2	Capabilities and Limitations of LLM Generation	6
2.3	The HLE Paradigm: A Case Study in Principled Benchmark Construction	7

3	Automated and Semi-Automated Generation Pipelines	8
3.1	Hypothesis-Driven Error Analysis Pipelines	9
3.2	Self-Play and Dual-Role Evaluation Frameworks	9
3.3	Multi-Agent Verification Pipelines	10
3.4	Automated Benchmark Auditing: BenchBench	10
3.5	Dynamic and Living Benchmarks	11
3.6	Prompt Engineering and Chain-of-Thought Extension	11
3.7	Adversarial and Iterative Refinement Mechanisms	11
3.8	Neuro-Symbolic and Formal Synthesis	12
4	Human-in-the-Loop Collaborative Generation	13
4.1	RLHF and Its Variants for Question Quality Optimization	13
4.2	Explicit Human-Machine Collaborative Frameworks	14
4.3	Multi-Agent Verification as Structural Necessity	14
4.4	Human Feedback as a Scaling Signal	15
5	Domain-Specific and Scientific Application Pipelines	16
5.1	Mathematics: From Theorem Proving to Competition Problems	17
5.2	Physics and Chemistry: Simulation-Based and Tool-Augmented Challenges	17
5.3	Materials Science and Knowledge Graph Integration	18
5.4	Biomedical Sciences and Genomics	18
5.5	Legal, Financial, and Actuarial Reasoning	19
5.6	Cross-Domain Patterns	19
6	Evaluation Systems and Benchmark Ecosystems	20
6.1	Multi-Dimensional Quality Assessment	21
6.2	The Contamination Crisis	21
6.3	Benchmark Ecosystems and Their Evolution	22
6.4	The Meta-Evaluation Problem	22
7	Future Directions and Open Challenges	22
7.1	Methodological Convergence and Hybrid Architectures	23
7.2	Multimodal Question Synthesis	23
7.3	Personalized and Adaptive Difficulty Calibration	23
7.4	Data Contamination and Evaluation Integrity	23
7.5	Ethical and Societal Considerations	24
7.6	Toward Autonomous Scientific Inquiry	24
8	Conclusion	24

1 Introduction

1.1 Background and Motivation

The rapid evolution of large language models (LLMs) [28]—from early GPT variants [26] through open-source families like LLaMA [25], Llama 3 [70], and Mixtral [71] to the latest generation of frontier models—has precipitated an acute evaluation crisis. Traditional static benchmarks, including MMLU [2], GSM8K [17], and MATH [18], are approaching saturation, with frontier models achieving near-ceiling performance [26, 27]. More critically, empirical contamination analysis by Song et al. [57] has demonstrated that benchmark scores may “conflate exam-oriented competence with principled capability,” with widespread above-baseline gains observed under noisy perturbation conditions across multiple models.

The emergence of “Humanity’s Last Exam” (HLE) [1] crystallized a community-wide recognition that evaluation must evolve from static testbeds to dynamic, contamination-resistant challenges demanding the full spectrum of human cognitive capabilities. Yet HLE’s construction—LLM-assisted screening followed by expert rewriting—is expensive and difficult to scale, motivating a fundamental question: *How can LLMs themselves be leveraged to synthesize assessment items of HLE-caliber difficulty at scale?*

This question sits at the intersection of two broader research movements. The first is the rapidly maturing field of *synthetic data generation for LLMs*, which has established that LLM-synthesized data can serve as a viable substitute for human-curated data across training, fine-tuning, and evaluation [59–61]. The second is the *LLM-as-agent* paradigm, in which LLMs are equipped with tools, planning capabilities, and multi-agent coordination to perform complex, multi-step tasks [31, 33, 34]. Difficult question synthesis represents a natural convergence of these movements: it requires both the data generation expertise of the first and the agentic orchestration of the second.

The significance of this convergence manifests across three interconnected dimensions. First, automated question synthesis enables perpetually evolving benchmarks that resist saturation. MathDuels [52] demonstrates this concretely through self-play: as newer models enter the arena, they produce problems that defeat previously dominant solvers, creating a co-evolutionary dynamic in which benchmark difficulty escalates in tandem with model capability. Second, in domain-specific contexts, question synthesis can serve as an “inspiration engine” for scientific discovery. DBench-Bio [55] transforms newly published research into testable hypothesis questions, while ChemCrow-inspired pipelines [14] identify unsolved synthesis bottlenecks as research questions, effectively converting gaps in the literature into evaluation challenges. Third, the educational implications are profound: the hypothesis-driven approach of Fu et al. [51] demonstrates that the most effective diagnostic questions precisely target specific conceptual blind spots, suggesting a path toward truly individualized assessment that adapts to each learner’s knowledge profile.

1.2 Scope and Research Questions

This survey addresses: *What are the state-of-the-art pipelines for synthesizing HLE-level or domain-specific high-difficulty questions using LLMs, how are they designed and implemented, and how is their output evaluated?*

To answer this, we examine the full pipeline from end-to-end workflows that transform input materials into output artifacts, through the methodological repertoire spanning prompt engineering, iterative refinement, curriculum learning, adversarial generation, and reinforcement learning, to the critical role

of human experts in guiding, filtering, and providing feedback at strategic junctures. We further consider domain-specific applications across mathematics, physics, chemistry, materials science, and beyond, alongside the evaluation criteria and benchmark ecosystems used to assess quality, difficulty, and novelty.

1.3 Relation to Existing Surveys

Several recent surveys provide complementary perspectives. Long et al. [59] survey LLM-driven synthetic data generation, curation, and evaluation across all data types, establishing a generic workflow framework. Tan et al. [60] focus specifically on LLM-based data annotation and synthesis, with emphasis on quality assessment. Wang et al. [61] examine data synthesis and augmentation across the LLM lifecycle from pre-training to alignment. The contamination perspective is surveyed separately [62], documenting how benchmark integrity is compromised.

Our survey differs from these works in a critical respect: we focus specifically on the synthesis of *difficult assessment items*—questions designed to probe the capability frontier rather than to augment training data. This specialization introduces unique challenges (difficulty calibration, novelty verification, solvability assurance) that are peripheral to general-purpose data synthesis. At the same time, we build on the foundational insights of these surveys, particularly regarding generation workflows, quality filtering, and the role of human feedback.

1.4 Contributions and Organization

Our contributions are fourfold. We present a systematic taxonomy of difficult question synthesis methods, organized along the automation–human-involvement spectrum, with particular emphasis on *how* questions are produced rather than *what* benchmarks exist. We provide critical analysis of concrete, implemented pipeline architectures with published experimental results—including hypothesis-driven error analysis [51], self-play evaluation [52], multi-agent verification [54], and automated benchmark auditing [53]—dissecting their internal mechanisms, failure modes, and empirical performance. We trace the methodological evolution from one-shot prompting through iterative self-refinement, adversarial self-play, and multi-agent verification to the current frontier of autonomous synthesis ecosystems (Figure 1), offering a unified perspective on the paradigm shift that defines the field. And we identify open challenges and articulate a research agenda for the field’s continued development.

The remainder is organized as follows. Section 2 establishes theoretical foundations. Section 3 surveys automated pipelines. Section 4 examines HITL paradigms. Section 5 discusses domain applications. Section 6 reviews evaluation systems. Section 7 identifies future directions. Section 8 concludes.

2 Theoretical Foundations and Core Challenges

2.1 Defining “Difficulty”: A Multi-Dimensional Framework

Before addressing the synthesis of difficult questions, a rigorous definition of “difficulty” is prerequisite. Drawing on the recent literature, we characterize high-difficulty assessment items along six orthogonal dimensions that, taken together, capture the multifaceted nature of genuine intellectual challenge.

The first dimension is **reasoning depth**. A truly difficult question demands multi-step, non-linear reasoning chains in which simple information retrieval or single-step logical deduction is insufficient. Empirical evidence from the MATH dataset and its derivatives demonstrates that physics and mathematics problems typically require substantially longer reasoning chains than other disciplines, and that chain-of-thought prompting yields the largest performance gains precisely on items requiring extended reasoning [6, 16, 18]. The second dimension, **knowledge breadth and integration**, captures the requirement that a question span multiple sub-fields or even disciplines, forcing the solver to flexibly invoke and synthesize heterogeneous knowledge sources. Cross-disciplinary integration remains particularly challenging for current LLMs, whose training data is often partitioned by domain [2]. The third dimension, **counter-intuitiveness and creativity**, characterizes questions whose optimal solution path is non-obvious or contrary to standard heuristic patterns. Such questions effectively probe a model’s capacity for “out-of-the-box” reasoning and have been shown to discriminate strongly between models that rely on pattern matching and those capable of genuine generalization [6, 19].

The fourth dimension, **informational imperfection**, reflects the reality that real-world scientific inquiry rarely begins with a perfectly specified problem. Questions containing redundancy, ambiguity, or deliberate omissions require the solver to filter, distill, and hypothesize—capabilities that mirror authentic research practice [47]. The fifth dimension, **novelty and contamination resistance**, is perhaps the most practically urgent: questions must be genuinely new, not amenable to retrieval from existing corpora or the web. This dimension is critical for maintaining benchmark integrity in the face of the data contamination that Song et al. [57] have empirically demonstrated to be widespread. The sixth and final dimension, **clarity and solvability**, imposes the constraint that despite its difficulty, a question must be well-posed: precisely defined, logically self-consistent, and possessing at least one verifiable solution. Ambiguity arising from poor construction is not legitimate difficulty but rather a failure of question design [1].

These six dimensions form a conceptual scaffold that underpins both the generation methodologies discussed in Section 3 and the evaluation criteria examined in Section 6. Before proceeding to the detailed pipeline analysis, Figure 1 provides a panoramic view of how the synthesis methodology itself has evolved, tracing the progression from primitive one-shot prompting to the complex, multi-agent creation ecosystems that define the current state of the art.

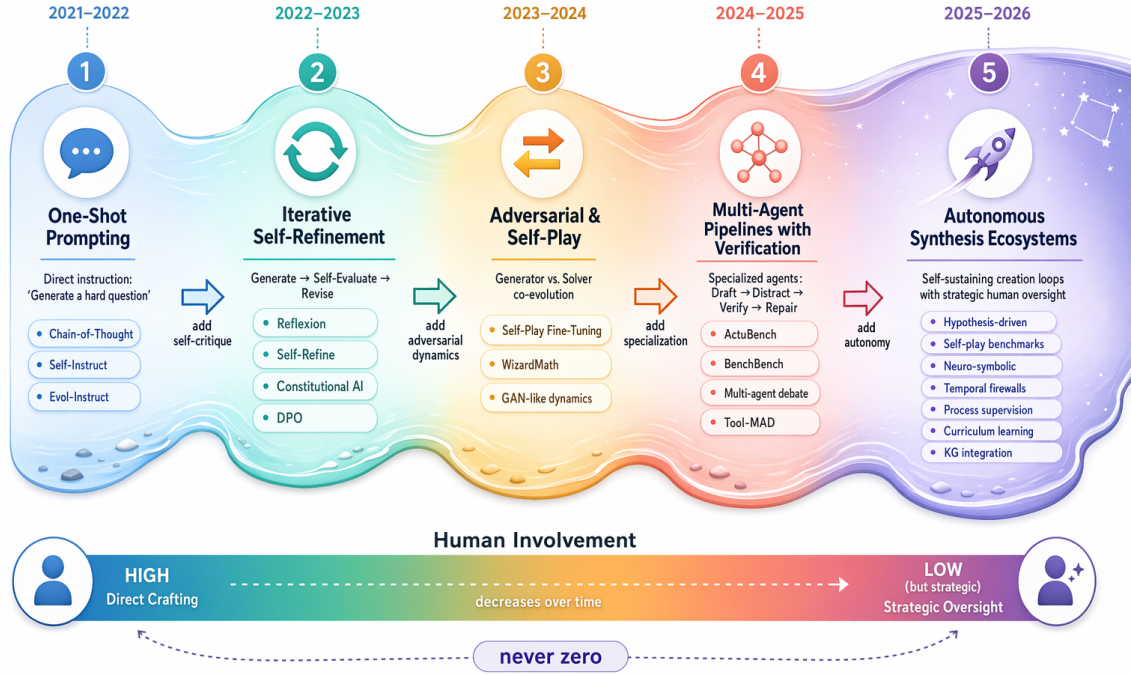


Figure 1: Methodological evolution of LLM-driven difficult question synthesis. The figure traces five paradigmatic stages through which synthesis pipelines have progressively matured. **Stage 1: One-Shot Prompting** (2021–2022)—the earliest approach directly instructs the LLM to generate a difficult question in a single pass, leveraging chain-of-thought prompting [6] and Self-Instruct [11] but lacking any quality assurance mechanism. **Stage 2: Iterative Self-Refinement** (2022–2023)—the introduction of generate-evaluate-revise loops through Reflexion [41], Self-Refine [42], and Constitutional AI [10] enables the LLM to critique and improve its own output, though self-evaluation remains unreliable for genuinely difficult items. **Stage 3: Adversarial and Self-Play** (2023–2024)—GAN-like dynamics and self-play fine-tuning [67] introduce co-evolutionary pressure between generator and solver agents, naturally escalating difficulty without human intervention. **Stage 4: Multi-Agent Pipelines with Verification** (2024–2025)—the decomposition of synthesis into specialized agent roles (drafting, distractor generation, independent verification, knowledge grounding), as exemplified by ActuBench [54] and BenchBench [53], establishes that verification is a structural necessity rather than an optional refinement. **Stage 5: Autonomous Synthesis Ecosystems** (2025–2026)—the current frontier integrates hypothesis-driven error analysis [51], self-play living benchmarks [52], neuro-symbolic formal guarantees [80], temporal contamination firewalls [55], process reward supervision [107], curriculum learning [83, 101], and knowledge graph integration [94–96] into self-sustaining creation loops that combine LLM scalability with strategic human oversight. The bottom bar indicates that human involvement transitions from high-volume direct crafting to concentrated strategic oversight, but never reaches zero.

2.2 Capabilities and Limitations of LLM Generation

LLMs have evolved from rudimentary text completion engines to systems capable of following complex multi-step instructions, generating structured outputs, and performing elementary logical reasoning. The Chain-of-Thought (CoT) paradigm [6] was a pivotal advance, demonstrating that explicitly eliciting intermediate reasoning steps substantially improves performance on complex tasks. Subsequent developments—including self-consistency decoding, tree-of-thought reasoning [65], graph-of-thought reasoning [64], self-discovered reasoning structures [63], and tool-augmented generation [32–34, 68]—

have further extended the frontier. The emergence of reasoning-specialized models such as OpenAI o1 [73] and DeepSeek-R1 [72], which employ reinforcement learning to elicit extended chain-of-thought reasoning at inference time, has demonstrated that scaling test-time compute [74] can yield dramatic improvements on mathematical and scientific benchmarks including GPQA [3], FrontierMath [4], and LiveBench [5]. The GRPO-VPS framework [107] extends this line of work by demonstrating that verifiable process supervision—step-level reward signals derived from formal verification—yields substantial gains over outcome-only supervision, establishing that the granularity of reward signals is a critical lever for reasoning capability. The process reward model literature [81, 110] has independently converged on a similar conclusion: generating precise, scalable training data for step-level rewards is essential for advancing multi-step reasoning capabilities. The Self-Instruct framework [11] demonstrated that LLMs can generate their own instruction-following training data, while WizardLM’s Evol-Instruct [12] showed that progressive instruction evolution can produce increasingly complex training examples. The “Textbooks Are All You Need” work [66] further demonstrated that synthetic textbook-quality data can train surprisingly capable models, establishing that data quality—not merely data quantity—is the critical lever in synthetic data pipelines. These foundational works established that LLM-based data synthesis is viable at scale, a prerequisite for the more specialized question synthesis pipelines surveyed in this paper.

Nevertheless, several intrinsic limitations constrain the direct application of LLMs to high-difficulty question synthesis. The hallucination problem—in which LLMs fabricate plausible but incorrect information—is particularly fatal for scientific question synthesis, where factual precision is non-negotiable [29]. LLMs also tend to reproduce problem patterns and solution templates common in training data, making it difficult to generate structurally novel challenges [20]. Naively instructing an LLM to “generate a difficult question” typically yields surface-level inflation—longer problem statements, obscure terminology, or computational complexity—rather than deeper conceptual challenge [21]. Perhaps most critically, an LLM has inherent difficulty judging whether its own output is genuinely novel, logically sound, or appropriately calibrated in difficulty [35].

These limitations collectively imply that one-shot, single-prompt generation is inadequate for producing HLE-caliber questions, motivating the multi-stage pipelines surveyed in subsequent sections. The broader synthetic data literature [59–61] has independently reached a similar conclusion: high-quality synthetic data requires iterative refinement, external validation, and often human oversight—principles that are amplified in the context of difficult question synthesis where the margin for error is exceptionally narrow.

2.3 The HLE Paradigm: A Case Study in Principled Benchmark Construction

The construction process of HLE itself [1] serves as an instructive exemplar of principled benchmark design. The HLE pipeline integrates automated and human components in a carefully staged workflow beginning with LLM-assisted initial screening, in which a powerful LLM identifies candidate questions from large corpora of academic resources, filtering for items that appear challenging and well-formulated. This is followed by expert review and rewriting, where domain specialists rigorously vet the candidates, discarding low-quality items and substantially rewriting promising ones to ensure scientific rigor, novelty, and appropriate difficulty calibration. Finally, all items undergo manual approval before inclusion in the benchmark.

This process crystallizes a central thesis that pervades the literature: at the current state of the art,

the creation of truly high-quality difficult questions is not a purely automated process but a synergistic product of LLM scalability and deep human expertise. This thesis provides the intellectual foundation for both the automated pipelines (Section 3) and HITL paradigms (Section 4).

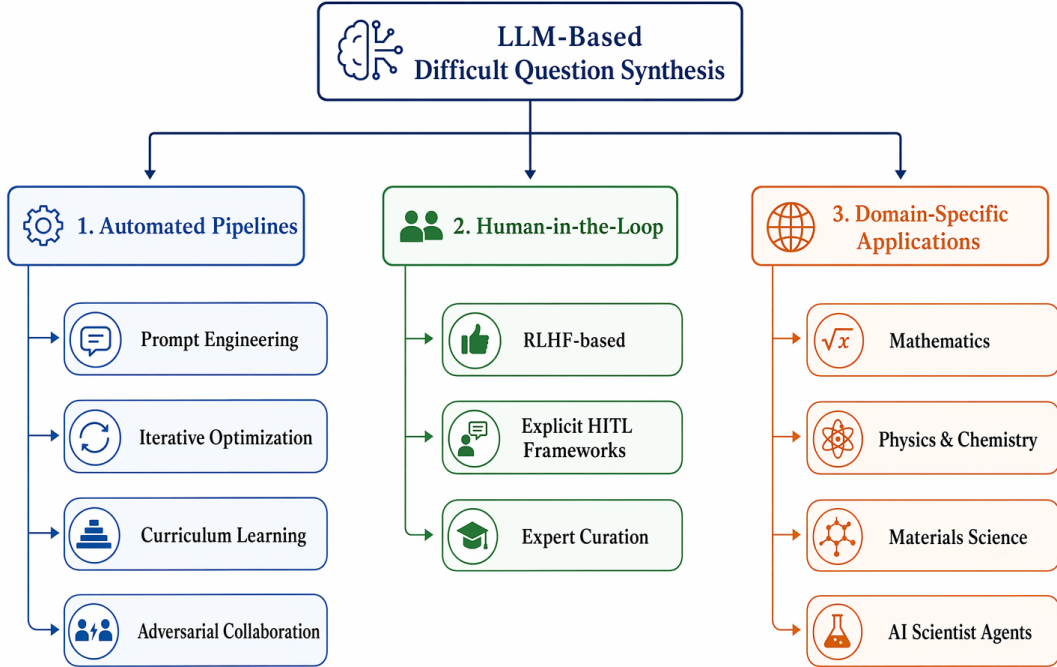


Figure 2: Taxonomy of LLM-based difficult question synthesis methods. The field is organized along three axes: automated pipelines—including prompt engineering (PromptCoT 2.0, reverse chain-of-thought), iterative optimization (self-refinement, Reflexion), self-play evaluation (MathDuels), hypothesis-driven generation (Fu et al.), benchmark auditing (BenchBench), dynamic benchmarks (DBench-Bio, LiveMathematicianBench, TimeSeriesExamAgent), and neuro-symbolic synthesis (CircuitSynth); human-in-the-loop paradigms—including RLHF/DPO variants, Constitutional AI, explicit HITL frameworks (exploration-exploitation, user-driven exploration), and multi-agent verification (ActuBench); and domain-specific applications—including mathematics (MathGenie, CHASE, formal verification), physical sciences (ChemCrow, DeepRetro), materials science (MatKG, MIND), biomedical sciences (DBench-Bio, PhageBench, ToxReason), legal and actuarial reasoning (ActuBench, NitiBench, TRIDENT), and AI scientist agents (DeepResearchEco).

3 Automated and Semi-Automated Generation Pipelines

This section provides a detailed analysis of concrete, implemented pipeline architectures for synthesizing high-difficulty assessment items. Rather than surveying abstract frameworks, we focus on systems with published experimental results, examining their internal mechanisms, failure modes, and empirical performance. The methodological progression traced in Figure 1 provides the organizing principle: we begin with hypothesis-driven and prompt-engineering approaches (Stages 1–2), proceed through self-play and adversarial mechanisms (Stage 3), examine multi-agent verification architectures (Stage 4), and conclude with the integrated synthesis ecosystems that define the current frontier (Stage 5).

3.1 Hypothesis-Driven Error Analysis Pipelines

A particularly compelling approach is the hypothesis-driven error analysis pipeline proposed by Fu et al. [51], which exemplifies a growing class of methods that treat benchmark generation not as an open-ended creative task but as a targeted diagnostic process. The pipeline operates in two principal stages. In the first stage, an LLM analyzes a target model’s error patterns on an existing benchmark such as MATH [18], formulating explicit hypotheses about which specific mathematical concepts or skills cause failures—for example, the system might hypothesize that a model systematically fails at problems requiring the Chinese Remainder Theorem in non-standard modular arithmetic settings. In the second stage, new benchmark problems are generated specifically to test the identified weaknesses, with the generation prompt incorporating the hypothesis as a conditioning signal.

The critical empirical finding is a strong positive correlation between hypothesis accuracy and generated problem difficulty: problems derived from the most accurate hypotheses reduce Llama-3.3-70B-Instruct’s accuracy from 77% on MATH to as low as 45%. This finding has two important implications. First, difficulty is not random—the most challenging problems are those that precisely target a model’s specific conceptual blind spots, not those that are generically “hard.” Second, error analysis is a first-class generation tool—the pipeline treats the target model’s failure patterns as the primary input to the generation process, establishing a paradigm in which benchmark creation is driven by diagnostic insight rather than difficulty heuristics. The authors further demonstrate that the pipeline generalizes beyond mathematics, suggesting broad applicability wherever LLM error patterns can be systematically characterized.

This diagnostic paradigm connects naturally to curriculum learning approaches that dynamically adjust synthesis difficulty. CoTEvol [82] introduces self-evolving chain-of-thoughts, wherein the reasoning traces used to generate mathematical problems are themselves iteratively refined through a co-evolutionary loop, producing progressively more sophisticated problem-solution pairs. MathMixup [83] demonstrates that difficulty-controllable data synthesis, when combined with curriculum learning, yields substantial gains in mathematical reasoning: the system generates problems at precisely calibrated difficulty levels and sequences them according to a curriculum schedule that mirrors pedagogical progression. The curriculum-aware budget scheduling framework of AvoidOverthink [101] addresses a complementary challenge: allocating inference compute across problems of varying difficulty to avoid both overthinking on easy items and underthinking on hard ones, demonstrating that adaptive resource allocation improves overall reasoning efficiency. ORACLE [84] takes a constraint-led approach, using explicit logical and semantic constraints to elicit higher-quality synthetic reasoning data from LLMs, while Socratic-Zero [85] demonstrates that reasoning capabilities can be bootstrapped through data-free agent co-evolution, eliminating the need for human-curated seed data entirely. The adaptive multi-expert reasoning framework [108] routes problems to specialized expert modules based on difficulty estimates and uncertainty quantification, achieving performance gains that scale with problem complexity.

3.2 Self-Play and Dual-Role Evaluation Frameworks

MathDuels [52] introduces a fundamentally different paradigm: a self-play benchmark in which models simultaneously serve as both problem posers and problem solvers. The key insight is that authoring and solving capabilities are partially decoupled—a model that excels at solving may produce mediocre problems, and vice versa. The pipeline employs a three-stage architecture comprising meta-prompting

for initial problem conception, problem generation for actual creation, and difficulty amplification for post-processing. An independent automated verifier excludes ill-posed questions.

A psychometrically principled innovation is the use of Rasch modeling [58] to jointly estimate solver abilities and problem difficulties from the observed solve/fail matrix, producing a continuous, interval-scale measure of difficulty that is more informative than raw solve rates. The most distinctive property of MathDuels is its living-benchmark character: as newer, more capable models enter the arena, they produce problems that defeat previously dominant solvers, creating a self-sustaining difficulty escalation. An evaluation of 19 frontier models demonstrated that dual-role evaluation reveals capability separations invisible in single-role benchmarks.

3.3 Multi-Agent Verification Pipelines

ActuBench [54] presents a multi-agent pipeline for generating actuarial examination items aligned with the IAA Education Syllabus. Its significance lies in the explicit demonstration that multi-agent verification is load-bearing—the verification step is not a cosmetic addition but a structural necessity. The four-agent architecture comprises a Drafting Agent that generates initial items, a Distractor Agent that constructs plausible wrong-answer options, a Verifier Agent that independently verifies both stages and drives bounded one-shot repair loops, and an Auxiliary Agent handling Wikipedia-note summarization and topic labeling. The verifier flags a majority of drafted items on first pass, underscoring that unverified LLM output is inadequate for high-stakes assessment. The study also found that MCQ scaffolding inflates the performance ceiling—MCQ and LLM-as-Judge rankings differ meaningfully—making open-ended evaluation with judge scoring essential for discriminating at the capability frontier.

Building on this multi-agent paradigm, Ikram et al. [76] introduce a framework for personalized math problem generation in which four specialized validator agents independently assess each generated problem against solvability, realism, readability, and authenticity. Evaluated on 600 problems from the ASSISTments platform, the study finds that authenticity and realism are the most common initial failure modes, but a single refinement iteration substantially reduces these failures. Walkington et al. [77] extend this with a teacher-in-the-loop study involving eight middle school math teachers who produced 212 personalized problems, finding that both teachers and students sought modifications to the fine-grained personalized elements of the real-world context—indicating that current LLMs struggle with the subtle cultural and contextual knowledge required for deep personalization.

3.4 Automated Benchmark Auditing: BenchBench

BenchBench [53] represents a meta-level contribution: rather than generating questions for a specific domain, it provides a framework for benchmarking the benchmark generation process itself. The three-stage pipeline comprises domain card extraction from existing seed benchmarks, LLM-based generation of quota-controlled suites, and validation using a multi-model answerer panel with exact/numeric/symbolic verifiers where possible and rubric-guided judging otherwise. Across nine variants spanning computer science, mathematics, medicine, and theory-of-mind reasoning, BenchBench generated 16.7K items, retaining approximately 15K post-filtering, and produced approximately 152K graded model-item responses. Two key empirical results emerge: benchmark-design ability is only moderately correlated with answer-time strength (Spearman $\rho \approx 0.37$), confirming that the ability to create difficult questions is a partially independent skill; and invalidity is negatively associated with discrimination (Pearson $r \approx 0.62$),

indicating that poorly constructed items are not merely uninformative but actively harmful to benchmark quality.

3.5 Dynamic and Living Benchmarks

DBench-Bio [55] addresses the contamination problem through a temporal firewall: because the benchmark is regenerated monthly from newly published paper abstracts, items cannot have appeared in any model’s training data. The three-stage pipeline—data acquisition from authoritative abstracts, LLM-based QA extraction, and quality filtering for relevance, clarity, and centrality—produces a continuously fresh evaluation set spanning 12 biomedical sub-domains. LiveMathematicianBench [75] employs a complementary strategy, grounding its problems in recent arXiv papers published after model training cutoffs and employing a thirteen-category logical taxonomy of theorem types with a proof-sketch-guided distractor pipeline. Under substitution-resistant evaluation, the best model achieves only 30.6% accuracy. Lee et al. [56] demonstrate that persona-based simulation can generate evaluation items that are more challenging than human-curated benchmarks, achieving linguistic and topical diversity comparable to human-curated datasets while being significantly harder for detection systems to identify. TimeSeriesExamAgent [78] extends the paradigm to time series reasoning, automatically generating benchmarks from real-world datasets spanning healthcare, finance, and weather, evaluating pattern recognition, noise understanding, similarity analysis, anomaly detection, and causality.

3.6 Prompt Engineering and Chain-of-Thought Extension

PromptCoT 2.0 [23] employs “task synthesis”—directing the LLM to generate not only questions but also accompanying difficulty rationales and solution sketches. The approach builds on the broader finding that mathematical reasoning capabilities can be substantially improved through targeted synthetic data generation, as demonstrated by DeepSeekMath [22] which showed that carefully curated mathematical pre-training data yields significant gains on competition-level benchmarks. The meta-prompt incorporates explicit directives to steer generation toward specific difficulty profiles, producing questions at AIME and HMMT competition levels. A complementary strategy, reverse chain-of-thought engineering, inverts the question-to-solution direction: the LLM first generates a complex reasoning chain, then constructs a problem for which this chain constitutes the optimal solution path, and finally obscures the cues that point toward the key steps. This “solution-to-question” inversion naturally produces problems with intricate structure and non-standard solution paths.

3.7 Adversarial and Iterative Refinement Mechanisms

Drawing on GAN-like dynamics [24], multi-agent adversarial collaboration systems pit a generator against a solver, with a critic providing natural-language feedback that progressively escalates difficulty. The self-play fine-tuning paradigm [67] demonstrates that a model’s own generated data can be used to iteratively strengthen both generation and evaluation capabilities, establishing a foundation for the co-evolutionary dynamics exploited by systems like MathDuels. Multi-agent orchestration frameworks such as MetaGPT [69] provide the architectural substrate for these adversarial systems, enabling specialized agents to collaborate through structured message passing. Self-refinement [41, 42]—in which the LLM generates, self-evaluates, and revises—provides a lightweight iterative mechanism that serves

as an effective first-pass filter when combined with external validation. The teach-LLMs-to-self-correct paradigm [43] extends this by training models to identify and repair their own reasoning errors, a capability that is directly applicable to question quality assurance. Self-correction research [106] further demonstrates that correction and refinement are distinct capabilities: genuine self-correction requires the model to identify errors in its own reasoning, not merely to refine surface-level features, a distinction that has direct implications for the reliability of automated question quality assurance. Pisano and Navigli [81] address a complementary challenge: generating precise, scalable training data for Process Reward Models by leveraging planning problems encoded in PDDL, producing approximately one million reasoning steps that yield substantial improvements in both mathematical and non-mathematical reasoning when used to augment existing PRM datasets. The GRPO-VPS framework [107] extends Group Relative Policy Optimization with verifiable process supervision, demonstrating that step-level reward signals derived from formal verification yield substantial gains over outcome-only supervision on competition-level mathematics. Process reward models integrated with planning [110] further demonstrate that generating precise step-level reward training data through PDDL-encoded planning problems can scale PRM datasets to approximately one million reasoning steps, with downstream improvements in both mathematical and code reasoning tasks.

3.8 Neuro-Symbolic and Formal Synthesis

Cheng et al. [80] introduce CircuitSynth, a neuro-symbolic framework that decouples semantic reasoning from surface realization by distilling a Teacher LLM’s reasoning capabilities into a Probabilistic Sentential Decision Diagram (PSDD). The PSDD structurally enforces hard logical constraints while allowing convex optimization for soft distributional goals, achieving 100% schema validity on complex logic puzzles where unconstrained baselines reach only 12.4%. This work demonstrates that formal guarantees can be combined with the expressivity of neural generation, a direction that holds particular promise for mathematical and scientific question synthesis where logical soundness is paramount.

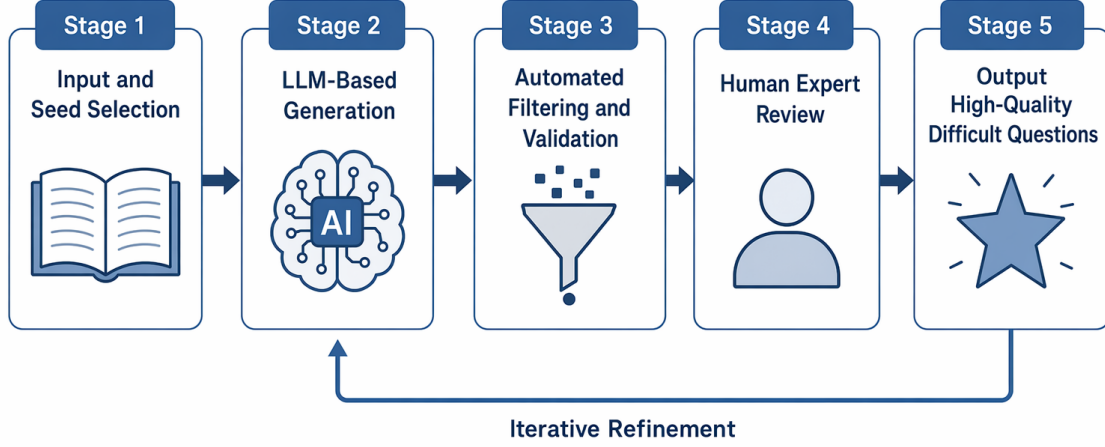


Figure 3: Multi-stage automated generation pipeline architecture. The pipeline flows from input and seed selection (knowledge bases, published literature, error pattern analysis) through LLM-based generation (PromptCoT 2.0, reverse chain-of-thought, Fu et al. hypothesis-driven, MathGenie back-translation, WizardMath Evol-Instruct, CoTEvol self-evolving chain-of-thought, ORACLE constrained elicitation), automated filtering and validation (BenchBench multi-model panel, ActuBench verifier agent, MathDuels Rasch modeling, DBench-Bio temporal firewalls, LiveMathematicianBench substitution-resistant evaluation, CircuitSynth formal verification, GRPO-VPS verifiable process supervision), and human expert review (HLE expert rewriting, Walkington et al. teacher-in-the-loop, Ikram et al. multi-agent validation), culminating in high-quality difficult questions. A feedback loop enables iterative refinement through self-refinement (Reflexion, Self-Refine), curriculum learning (MathMixup, AvoidOverthink), and self-correction.

4 Human-in-the-Loop Collaborative Generation

Purely automated pipelines, despite their scalability, often fall short when confronting challenges that demand extreme creativity, deep domain insight, or nuanced value judgments. The integration of human expertise into the generation loop—human-in-the-loop (HITL)—has consequently emerged as one of the most important and actively investigated paradigms in the field.

4.1 RLHF and Its Variants for Question Quality Optimization

Reinforcement Learning from Human Feedback (RLHF) [7, 8] was originally developed to align LLM outputs with human values of helpfulness, honesty, and harmlessness. Its core architecture—supervised fine-tuning followed by reward model training and reinforcement learning optimization—can be repurposed with minimal modification for question quality optimization. The canonical RLHF workflow comprises three stages: supervised fine-tuning on curated instruction-response pairs, reward model training from human preference rankings, and reinforcement learning optimization using algorithms such as PPO.

The adaptation for difficult question synthesis involves three principled modifications. First, the instruction is redesigned from “answer this question” to “generate a difficult question in [specified domain].” Second, human experts rank generated questions rather than answers, applying criteria specific to the six-dimensional difficulty framework: reasoning depth, knowledge integration, counter-intuitiveness, informational imperfection, novelty, and clarity. Third, the reward model learns to predict a composite “question quality score” across these dimensions, enabling the RL optimization loop to drive the generator toward increasingly sophisticated output. This adaptation effectively quantifies the tacit knowledge and aesthetic sensibility of domain experts into an optimizable reward signal. The Direct Preference Optimization (DPO) variant [9], which eliminates the separate reward model by directly optimizing the language model on preference data, offers a more streamlined alternative. Constitutional AI [10] further demonstrates that AI-generated feedback can substitute for human feedback in certain alignment contexts, suggesting a path toward scaling HITL beyond the bottleneck of human annotation capacity.

4.2 Explicit Human-Machine Collaborative Frameworks

Beyond RLHF’s implicit feedback mechanism, a distinct line of research focuses on constructing explicit interaction frameworks that position human experts as active collaborators at critical junctures of the generation pipeline. The exploration-exploitation paradigm [30] operates through a closed loop in which the LLM generates a diverse pool of candidate questions during the exploration phase, and domain experts then select the most promising candidates, decide which items are ready for the question bank or require further refinement, and provide specific natural-language instructions for revision. Expert feedback is appended to the prompt for the next exploration round, guiding the LLM toward progressively higher quality. The closed-loop structure ensures that human expertise is concentrated where it adds the most value—selection and high-level guidance—while the LLM handles the labor-intensive task of candidate generation.

An alternative HITL paradigm places the human expert in the role of an active explorer navigating a knowledge space, with the LLM serving as an intelligent assistant [49]. In a question synthesis context, the expert provides an initial concept or keyword, the LLM generates a constellation of related concepts and preliminary question ideas, the expert navigates by selecting nodes of interest and issuing targeted directives, and the LLM incorporates the expert’s exploration history to generate increasingly focused questions. This paradigm is particularly effective for interdisciplinary questions, where the expert’s ability to identify unexpected connections between fields is difficult to automate.

Frameworks designed for generating real-world challenges [35] demand especially deep human participation, since the constraints and complexities of real-world scenarios resist formal specification. A representative pipeline begins with a human expert defining a complex, open-ended scenario, continues with the LLM decomposing the macro-scenario into sub-problems and relevant variables, and culminates in iterative dialogue through which the expert and LLM refine, merge, and restructure the sub-problems into well-structured challenges incorporating authentic real-world constraints.

4.3 Multi-Agent Verification as Structural Necessity

A key empirical finding from the recent literature is that verification is not a cosmetic addition to generation pipelines but a structural necessity. ActuBench [54] provides the clearest evidence: its verifier agent flags a majority of drafted items on first pass, and without verification, the pipeline produces items with

unacceptable rates of ambiguity, unsolvability, and domain error. The ActuBench architecture—drafting, distractor generation, independent verification, and auxiliary knowledge retrieval—demonstrates a broader principle: specialization of agent roles improves quality. Each agent in the pipeline performs a distinct cognitive function (creative generation, adversarial option construction, critical evaluation, knowledge grounding), and the system’s output quality depends on the integrity of each role.

BenchBench [53] independently confirms that invalidity is strongly negatively associated with discrimination ($r \approx 0.62$), meaning that unverified items are not merely uninformative but actively degrade benchmark quality. The implication is clear: any serious question synthesis pipeline must incorporate independent verification, and the cost of verification is justified by the disproportionate damage caused by invalid items.

The multi-agent debate paradigm provides a complementary verification mechanism. Du et al. [86] demonstrate that diversity of thought—ensuring debating agents hold genuinely different perspectives—is the critical ingredient: homogeneous agents converge prematurely, while diverse agents surface errors and inconsistencies that would otherwise pass undetected. This finding has direct implications for question synthesis pipelines, where multiple independent evaluators with different reasoning strategies can catch a broader range of quality failures. Feng et al. [87] extend the multi-agent debate framework to fact-checking, demonstrating that structured debate among LLM agents produces more faithful explanations than single-agent approaches. Tool-MAD [88] integrates diverse tool augmentation with adaptive retrieval into the debate framework, enabling agents to ground their arguments in external evidence rather than relying solely on parametric knowledge. MAD-Fact [89] applies multi-agent debate specifically to long-form factuality evaluation, addressing the challenge that factual errors in extended generations are often subtle and context-dependent. The self-play debate paradigm [113] demonstrates that language models can improve the faithfulness and factual consistency of their own outputs through structured adversarial interaction, without external supervision. These multi-agent verification architectures collectively establish that specialization of cognitive roles—creative generation, adversarial critique, evidence grounding, and independent verification—is essential for maintaining quality in automated synthesis pipelines.

4.4 Human Feedback as a Scaling Signal

A critical insight emerging from the recent literature is that human feedback functions not merely as a quality filter but as a scaling signal for the entire generation ecosystem [13]. As LLMs become more capable, the human expert’s role shifts from detailed crafting to strategic oversight—defining evaluation criteria, calibrating difficulty thresholds, and adjudicating edge cases. This evolution parallels the “weak-to-strong generalization” paradigm [13], where weaker supervisory signals are used to elicit capabilities that exceed the supervisor’s own level. Walkington et al.’s teacher-in-the-loop study [77] provides empirical grounding for this insight: teachers did not rewrite problems from scratch but rather provided targeted feedback on specific elements, suggesting that even modest human input can substantially improve LLM output when directed at the right junctures.

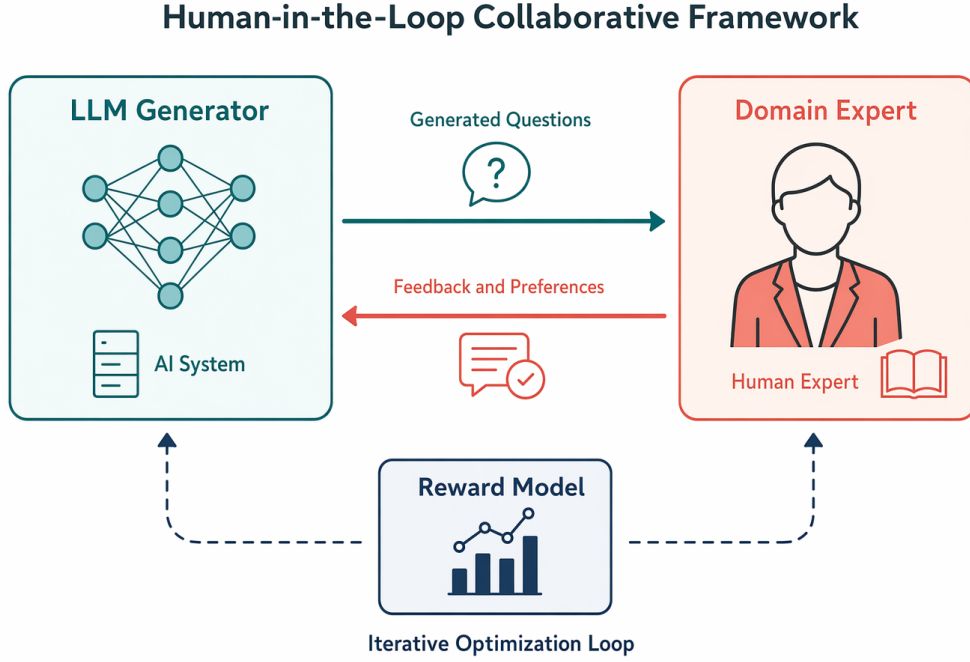


Figure 4: Human-in-the-loop collaborative generation framework. The LLM generator produces candidate questions, which are evaluated by domain experts through multiple feedback channels. Expert feedback—including selection, textual revision instructions, quality ratings, and preference rankings—feeds back into the generation loop, progressively elevating question quality. Representative implementations: RLHF/DPO (Ouyang et al., Rafailov et al.) for implicit preference alignment; Constitutional AI (Bai et al.) for AI-generated feedback substitution; ActuBench (Schmidt et al.) multi-agent verification with verifier flagging majority of items on first pass; Walkington et al. teacher-in-the-loop for personalized problem refinement; Ikram et al. four-agent validation against solvability, realism, readability, and authenticity; MathDuels (Xu et al.) self-play for co-evolutionary difficulty escalation; BenchBench (Zheng et al.) meta-evaluation confirming that invalidity strongly degrades discrimination ($r \approx 0.62$); multi-agent debate frameworks (Du et al. diversity of thought, Feng et al. faithful fact-checking, Tool-MAD tool-augmented verification, MAD-Fact long-form factuality, self-play debate for faithfulness); and Socratic-Zero data-free agent co-evolution for bootstrapping reasoning capabilities.

5 Domain-Specific and Scientific Application Pipelines

The application of general-purpose generation pipelines to specific scientific domains requires deep integration with domain-specific knowledge representations, computational tools, and methodological conventions. The broader landscape of LLM applications in science [38] provides essential context: domain-specific question synthesis is a specialized instance of the general challenge of equipping LLMs with scientific reasoning capabilities. Autonomous research systems such as DeepResearchEco [39] represent the most ambitious realization of this vision, orchestrating literature search, hypothesis generation, experimental design, and result interpretation in self-sustaining research loops. This section examines the most mature application areas, highlighting both the common patterns and the domain-specific innovations that distinguish each field’s approach to question synthesis.

5.1 Mathematics: From Theorem Proving to Competition Problems

Mathematical question generation has been a focal point for LLM-based synthesis, owing to the formal verifiability of mathematical statements and the existence of well-defined difficulty hierarchies. Beyond the PromptCoT 2.0 pipeline and MathDuels self-play framework discussed in Section 3, several specialized systems target mathematical problem generation with distinct methodological innovations. MathGenie [45] employs a question back-translation strategy in which the LLM first generates a solution and then constructs a problem that admits that solution, while systems like CHASE [46] focus on formal verification of generated problems using proof assistants.

A canonical mathematical question generation pipeline proceeds through five stages: concept or theorem selection from a structured knowledge base, structural generation of a logical proof or computation skeleton, numerical and symbolic instantiation grounding the abstract structure in a specific problem instance, natural language packaging translating the formal problem into fluent prose, and verification using symbolic computation tools or formal proof assistants [44]. The integration of formal verification is a distinguishing feature of mathematical question synthesis, providing an objective, automated guarantee of correctness unavailable in other domains.

The personalized math problem generation work of Ikram et al. [76] and Walkington et al. [77] extends mathematical synthesis into educational applications, demonstrating that multi-agent validation frameworks can produce problems tailored to individual student interests while maintaining mathematical rigor. LiveMathematicianBench [75] pushes the frontier further by grounding problems in recent arXiv papers and employing proof-sketch-guided distractors that test genuine understanding rather than pattern matching.

5.2 Physics and Chemistry: Simulation-Based and Tool-Augmented Challenges

Physical science questions often require coupling theoretical reasoning with experimental or computational evidence. State-of-the-art generation pipelines integrate external simulation and calculation tools, beginning with the LLM defining a physical system or chemical reaction, generating input scripts for simulators such as COMSOL or quantum chemistry packages like Gaussian, running the simulation to obtain output data, and then generating an inverse problem requiring the solver to infer unknown parameters from the observed results. Such questions test not only theoretical knowledge but also data analysis and modeling skills, closely mirroring authentic scientific practice.

ChemCrow [14] represents the most mature tool-augmented synthesis pipeline in chemistry. By integrating molecular editors, safety databases, and literature search tools, ChemCrow can autonomously plan synthesis routes and, crucially, identify bottlenecks where no satisfactory solution exists in the literature. These bottlenecks are automatically formulated as high-quality research questions. DeepRetro extends this paradigm with iterative LLM reasoning for retrosynthetic pathway discovery, while Atom-anchored LLMs demonstrate domain-specific molecular representations that improve chemical reasoning accuracy. Recent work on bridging the plausibility-validity gap through fine-tuning reasoning-enhanced LLMs for chemical synthesis further illustrates the field’s maturation.

5.3 Materials Science and Knowledge Graph Integration

Knowledge graphs store entities and relationships in a structured format, providing LLMs with a robust factual foundation that mitigates hallucination. In materials science, the synergy between LLMs and KGs [36, 48, 49] offers a powerful substrate for generating scientifically meaningful questions. A materials science KG such as MatKG [37] contains entities including materials, properties, synthesis methods, and characterization techniques linked by typed relationships, enabling the LLM to identify missing links between materials and applications, contradictory relations arising from aggregating multiple literature sources, and unexplored paths connecting distant concepts.

Recent work on “From Data to Theory” agents for materials science and the MIND AI Co-Scientist for Material Research [94] demonstrates the rapid evolution of this field toward fully autonomous scientific discovery pipelines. GraphAgents [95] extends this paradigm by using knowledge graph-guided agentic AI for cross-domain materials design, orchestrating multiple specialized agents to traverse the KG, identify promising material candidates, and generate evaluation challenges that test whether models can discover non-obvious material-property relationships. HalluMat [97] addresses a critical quality assurance challenge specific to materials science: detecting hallucinations in LLM-generated materials content through multi-stage verification against authoritative knowledge bases, establishing that domain-specific hallucination detection is essential for maintaining scientific rigor in generated assessment items. KARMA [111] leverages multi-agent LLMs for automated knowledge graph enrichment, identifying missing links and contradictory relationships that can be formulated as research questions. The Discovery Engine [112] provides a broader framework for AI-driven synthesis and navigation of scientific knowledge landscapes, demonstrating how structured knowledge representations can guide the generation of scientifically meaningful evaluation challenges. HypoChainer [96] combines LLMs with knowledge graphs specifically for hypothesis-driven scientific discovery, creating a collaborative system that generates testable hypotheses by identifying gaps in existing knowledge graphs and formulating them as evaluation questions.

5.4 Biomedical Sciences and Genomics

The biomedical sciences present unique synthesis challenges due to the complexity of biological systems and the rapid pace of discovery. DBench-Bio [55] has already been discussed in the context of dynamic benchmarks. Complementing this work, PhageBench and GenomeQA test LLMs on raw genomic sequence understanding, while ToxReason benchmarks mechanistic chemical toxicity reasoning via adverse outcome pathways. These domain-specific benchmarks demonstrate that biological reasoning requires specialized evaluation frameworks that cannot be subsumed by general-purpose benchmarks.

Huang et al. [79] introduce a synthetic data generation pipeline for Objective Structured Clinical Examinations in French, combining controlled dialogue generation with LLM-assisted silver-labeling. The pipeline generates doctor-patient interview transcripts guided by scenario-specific evaluation criteria, combining ideal and perturbed performances to simulate varying student skill levels. MedPRM-Bench [90] introduces a fine-grained benchmark specifically for evaluating process reward models in medical reasoning, demonstrating that step-level supervision is critical for clinical question quality assessment and that existing PRMs exhibit systematic failures on medical reasoning chains. MR-Bench [91] provides a comprehensive survey and benchmark for medical reasoning with LLMs, establishing a taxonomy of clinical reasoning capabilities and identifying systematic gaps in current models’ diagnostic abil-

ities. EuropeMedQA [92] extends medical evaluation to multilingual and multimodal settings, demonstrating that clinical reasoning benchmarks must account for linguistic and cultural variation in medical practice. The emergence of “Towards a Medical AI Scientist” and “Towards Autonomous Mechanistic Reasoning in Virtual Cells” further signals the field’s trajectory toward fully autonomous biomedical research pipelines.

5.5 Legal, Financial, and Actuarial Reasoning

SynLexLM demonstrates that synthetic data combined with curriculum learning can scale legal LLMs, while domain-partitioned hybrid RAG architectures are being developed for legal reasoning in specific jurisdictions. The legal domain exemplifies a broader pattern: domain-specific synthesis requires not only factual accuracy but also sensitivity to jurisdictional, procedural, and interpretive nuances. ActuBench [54], already discussed in the context of multi-agent verification, represents one of the most mature domain-specific synthesis pipelines, with its alignment with the IAA Education Syllabus and dual MCQ/open-ended evaluation design providing a template for domain-specific benchmark construction. TimeSeriesExamAgent [78] extends the paradigm to financial and weather domain reasoning.

The legal domain has witnessed a proliferation of specialized benchmarks that illustrate the challenges of domain-specific question synthesis. PLawBench [93] introduces rubric-based evaluation for real-world legal practice, demonstrating that legal question synthesis requires not only factual accuracy but also sensitivity to the rubric structures used in actual legal assessment. BenGER [114] provides a collaborative benchmarking platform for German legal tasks, while the Korean Canonical Legal Benchmark [115] establishes knowledge-independent evaluation of legal reasoning capabilities, demonstrating that legal question synthesis must disentangle domain knowledge from reasoning ability. DeonticBench [104] targets reasoning over rules—a core legal competency—providing a benchmark that tests whether models can correctly apply deontic logic (obligations, permissions, prohibitions) to novel scenarios. Knowledge graph-assisted post-training for legal reasoning [117] demonstrates that structured legal knowledge representations can improve both the quality of generated legal questions and the accuracy of legal reasoning. A comprehensive survey of LLM evaluation in legal applications [105] catalogs the methodological challenges specific to legal question synthesis, including jurisdictional specificity, temporal validity of legal rules, and the distinction between black-letter law and interpretive reasoning. In chemistry, ChemGraph [98] provides an agentic framework for computational chemistry workflows that can identify unsolved problems in synthesis planning and formulate them as evaluation challenges. RxnNano [116] demonstrates that hierarchical curriculum learning can train compact LLMs for chemical reaction and retrosynthesis prediction, establishing that curriculum-based approaches extend naturally to chemical domain synthesis.

5.6 Cross-Domain Patterns

Across these diverse domains, several common patterns emerge. Tool integration is essential: mathematics requires formal verifiers, chemistry requires molecular editors and safety databases, physics requires simulators, biology requires genomic databases, and legal reasoning requires jurisdiction-specific knowledge bases. Domain-specific evaluation criteria matter: generic quality metrics are insufficient, and each domain requires specialized validators. Contamination resistance requires domain-specific strategies: formal verification in mathematics, temporal firewalls in biology, jurisdictional specificity in legal rea-

soning. And human expertise remains irreplaceable at the quality frontier: across all domains, the most valuable and hardest-to-automate contribution is the human expert’s ability to judge whether a generated question is not merely correct but interesting, insightful, and scientifically productive.

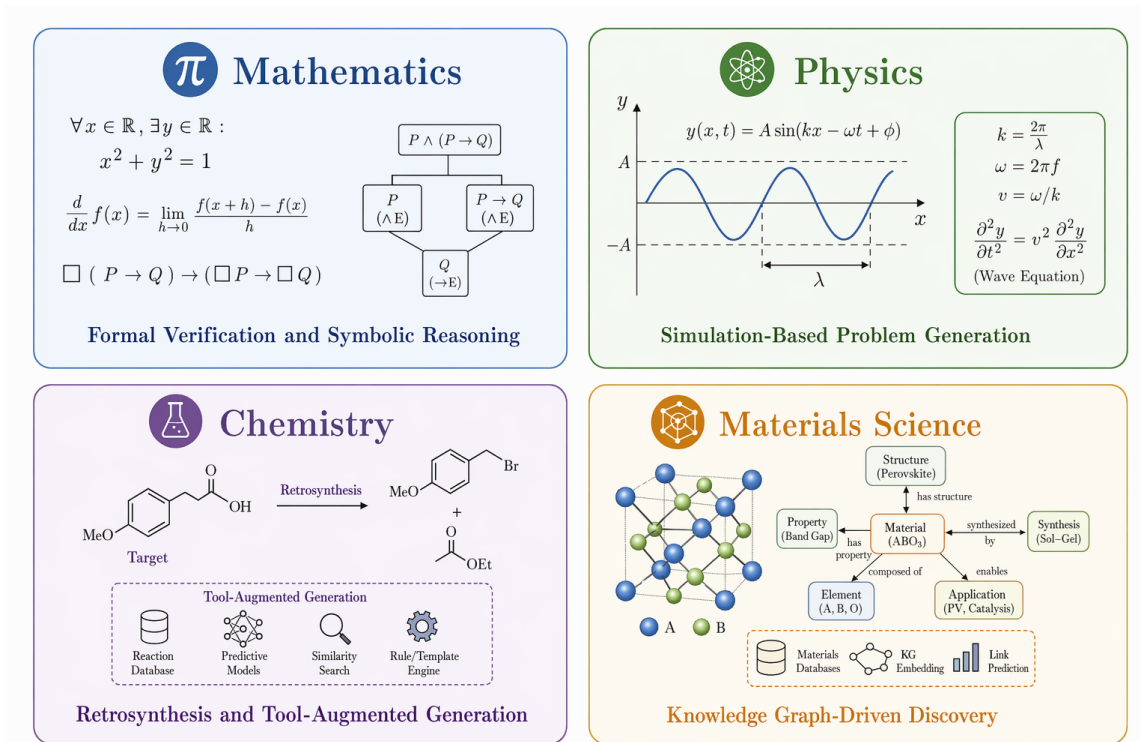


Figure 5: Domain-specific application pipelines for difficult question synthesis. Each quadrant represents a distinct domain: **mathematics**—formal verification via Isabelle and proof assistants, PromptCoT 2.0 for competition-level problems, MathGenie with question back-translation, CHASE for formal verification, LiveMathematicianBench with substitution-resistant evaluation, personalized generation (Ikram et al., Walkington et al.); **physics and chemistry**—simulation-based problem generation with COMSOL/Gaussian, ChemCrow for retrosynthetic route planning and bottleneck identification, DeepRetro for iterative retrosynthesis, Atom-anchored LLMs for molecular representations, ToxReason for mechanistic toxicity reasoning; **materials science**—knowledge graph integration via MatKG, MIND AI Co-Scientist, multi-agent high-throughput screening, “From Data to Theory” agents; **biomedical sciences**—DBench-Bio with monthly temporal firewalls, PhageBench and GenomeQA for genomic reasoning, Huang et al. OSCE pipeline for clinical assessment, MedPRMBench for process reward models, SurgCoT for surgical reasoning; **legal and financial reasoning**—ActuBench aligned with IAA Syllabus, NitiBench for Thai legal QA, TRIDENT for multi-domain safety, TimeSeriesExamAgent for temporal reasoning. Cross-domain patterns include tool integration, domain-specific validation, contamination resistance, and human expert oversight at the quality frontier.

6 Evaluation Systems and Benchmark Ecosystems

The generation of difficult questions is inseparable from the evaluation of their quality. This section examines the metrics, benchmarks, contamination dynamics, and meta-evaluation challenges that define the evaluation landscape, with emphasis on concrete empirical findings.

6.1 Multi-Dimensional Quality Assessment

Evaluating generated questions is a multi-faceted task that requires both automated and human assessment [35]. Automated metrics provide scalable first-pass filtering. Solvability analysis submits generated questions to one or more powerful solver LLMs, approximating difficulty by solve rate, time-to-solution, and reasoning trace complexity. MathDuels [52] formalizes this through Rasch modeling, producing continuous difficulty estimates rather than binary pass/fail labels. Textual complexity metrics—readability indices, lexical diversity, syntactic complexity—serve as coarse but inexpensive filters. Semantic similarity against existing question banks using embedding cosine similarity provides a contamination proxy, though this approach cannot detect more subtle forms of contamination where the underlying concept rather than the specific question has leaked into training data [57].

The gold standard remains expert evaluation along the six dimensions identified in Section 2: reasoning depth, knowledge integration, counter-intuitiveness, informational imperfection, novelty, and clarity. ActuBench [54] provides empirical evidence that MCQ scaffolding and LLM-as-Judge rankings differ meaningfully—MCQ evaluation inflates the performance ceiling, making judge-based scoring essential for discriminating at the frontier.

The use of LLMs as automated evaluators has become ubiquitous, but its reliability remains contested [35]. Recent work reveals several systematic failure modes including position bias (favoring responses presented first or last), verbosity bias (longer responses receive higher ratings regardless of accuracy), self-preference (favoring outputs from the same model family), and prompt sensitivity (minor variations in evaluation prompts produce substantially different rankings). BenchBench’s finding that benchmark-design ability is only moderately correlated with answer-time strength ($\rho \approx 0.37$) further complicates the use of a single model as both generator and evaluator.

6.2 The Contamination Crisis

Song et al. [57] provide the most systematic empirical analysis of benchmark contamination to date. Their audit framework employs a router-worker architecture comparing a clean-control condition against noisy conditions where benchmark problems are systematically deleted, rewritten, and perturbed. The core diagnostic: for a genuinely clean benchmark, noisy variants should not consistently outperform the baseline. The results are alarming: across multiple models, widespread but heterogeneous above-baseline gains were found under noisy conditions, indicating that benchmark-related cues may be reassembled and reactivate contamination-related memory. This finding challenges the assumption that benchmark scores directly reflect genuine generalization, suggesting instead that they may conflate exam-oriented competence with principled capability.

The literature converges on several design principles for contamination resistance. Temporal firewalls—regenerating benchmarks from sources published after model training cutoffs—represent the most rigorous approach, as demonstrated by DBench-Bio [55] and LiveMathematicianBench [75]. Dynamic generation through self-play, as in MathDuels [52], creates naturally escalating difficulty without relying on a fixed item pool. Provenance tracking through designer-answerer matrices, as in BenchBench [53], provides an audit trail linking each item to its generator, verifier, and grading outcome. Adversarial generation itself serves as a contamination-resistance mechanism, as demonstrated by Lee et al. [56].

6.3 Benchmark Ecosystems and Their Evolution

The classical benchmarks—MATH [18], GSM8K [17], SCIBENCH [16], and its successor SciBench 2 [50]—remain important reference points but face existential pressure. DynaMath [40] attempts to revitalize these by continuously incorporating newly generated, human-validated items. However, Song et al.’s contamination analysis [57] suggests that even “refreshed” benchmarks may retain contamination artifacts from their predecessors. HLE [1] represents the current frontier, but its construction—LLM screening followed by expert rewriting—is expensive and difficult to scale. The natural evolution is toward partial automation using the pipelines surveyed in Section 3, creating a virtuous cycle in which model evolution drives question synthesis, which drives harder benchmarks, which drives further model evolution.

The proliferation of domain-specific benchmarks—ActuBench for actuarial science, DBench-Bio for biomedicine, LiveMathematicianBench for mathematical reasoning, TimeSeriesExamAgent for time series, ToxReason for chemical toxicity—reflects a recognition that general-purpose benchmarks cannot adequately evaluate specialized capabilities. These domain benchmarks typically require deeper integration with field-specific knowledge representations and validation tools. Recent additions to this ecosystem include MedPRMBench [90] for medical process reward models, MR-Bench [91] for comprehensive medical reasoning evaluation, EuropeMedQA [92] for multilingual clinical assessment, PLawBench [93] for real-world legal practice, DeonticBench [104] for deontic rule reasoning, OptiVerse [103] for optimization problem solving, and OMIBench [109] for Olympiad-level multi-image reasoning. Auto-Bench [99] provides an automated benchmark specifically for evaluating scientific discovery capabilities in LLMs, while the Chomsky Hierarchy benchmark [102] evaluates formal reasoning capabilities across different levels of computational complexity, demonstrating that current LLMs exhibit systematic degradation as formal language complexity increases. The scientific hypothesis generation and validation survey [100] catalogs the methods, datasets, and evaluation frameworks used to assess whether LLMs can generate genuinely novel and testable scientific hypotheses.

6.4 The Meta-Evaluation Problem

BenchBench [53] raises a fundamental meta-evaluation question: who evaluates the evaluators? The finding that benchmark-design ability and solving ability are only moderately correlated implies that high-performing models are not necessarily good benchmark designers. Similarly, the finding that invalidity is strongly negatively associated with discrimination implies that quality control is not merely desirable but essential—bad items are worse than no items. Song et al. [57] advocate supplementing benchmark scores with explicit audits of contamination sensitivity and score confidence, moving from single-number metrics to multidimensional quality profiles. This represents a paradigm shift from “how well does the model perform?” to “how much should we trust this performance assessment?”

7 Future Directions and Open Challenges

The field of LLM-based difficult question synthesis is evolving rapidly. Several directions are likely to define the research agenda over the coming years.

7.1 Methodological Convergence and Hybrid Architectures

The most promising trajectory is the integration of multiple methodologies into unified, multi-layered systems. A prototypical architecture of the near future might combine a curriculum learning substrate that enables the generator model to progressively master question synthesis from simple to complex, an adversarial collaboration mechanism at the intermediate layer employing internal generator-solver competition to push difficulty and novelty, a human-in-the-loop overlay at the top tier where domain experts provide strategic oversight, and an RLHF-based feedback loop that internalizes expert judgments as a continuous training signal. Such hybrid architectures promise to combine the scalability of automation with the depth of human expertise, addressing the fundamental tension between quantity and quality.

7.2 Multimodal Question Synthesis

Real-world scientific challenges are rarely confined to text. Future systems must generate questions that integrate diagrams, images, experimental data, code, and interactive simulations [1]. This requires multimodal generation capabilities—LLMs with integrated vision and code generation modules capable of producing figures with subtle errors or hidden information—as well as cross-modal reasoning assessment frameworks and seamless coupling with drawing, simulation, and visualization tools.

7.3 Personalized and Adaptive Difficulty Calibration

The next generation of synthesis systems will generate questions dynamically calibrated to individual users’ knowledge profiles and cognitive capacities. In educational applications, this means generating exercises at each student’s zone of proximal development, as demonstrated by the personalized math problem generation work of Ikram et al. [76] and Walkington et al. [77]. In research applications, it means analyzing a researcher’s recent publications and stated interests to generate questions that optimally challenge their knowledge boundaries. Realizing this vision requires robust user modeling, fine-grained difficulty control, and careful attention to fairness and equity. The curriculum-aware budget scheduling framework [101] demonstrates that adaptive resource allocation—dynamically adjusting inference compute based on problem difficulty—improves overall reasoning efficiency, suggesting that future synthesis systems can co-optimize both question difficulty and computational resource allocation. The adaptive multi-expert reasoning paradigm [108] further illustrates how difficulty-aware routing can be integrated into question generation, with specialized expert modules handling different difficulty tiers.

7.4 Data Contamination and Evaluation Integrity

As generated benchmarks become more prominent, the contamination problem becomes more acute. Song et al.’s [57] empirical demonstration of widespread contamination sensitivity across multiple models reveals that the evaluation crisis is deeper than previously understood. Future research must address secure evaluation environments, continuous generation pipelines that produce fresh validated items on demand, and cryptographic or statistical techniques for provenance tracking and watermarking.

7.5 Ethical and Societal Considerations

The power to generate high-difficulty assessment items at scale carries significant ethical implications [1]. If high-quality personalized question generation becomes a premium commercial service, it may exacerbate existing inequalities in educational access. The technology could be weaponized to create unfair or biased examinations for gatekeeping purposes. Systematic biases in generation pipelines could produce large volumes of plausible but misleading research questions, wasting scientific resources. When AI-generated questions are used in high-stakes assessments, questions of responsibility and transparency become paramount. These risks underscore the necessity of establishing ethical guidelines, transparency requirements, and risk monitoring mechanisms commensurate with the technology’s capabilities.

7.6 Toward Autonomous Scientific Inquiry

Perhaps the most ambitious vision is the evolution from question synthesis to autonomous scientific inquiry—systems that not only generate questions but also pursue their answers, evaluate the results, and formulate new questions in a self-sustaining research loop [15]. The AI scientist agents discussed in Section 5 represent early steps toward this goal. HypoChainer [96] demonstrates the viability of hypothesis-driven discovery by combining LLMs with knowledge graphs to generate and validate scientific hypotheses, while the Discovery Engine [112] provides a broader framework for AI-driven synthesis and navigation of scientific knowledge landscapes. Scientific hypothesis generation and validation [100] catalogs the current methodological landscape and identifies key open challenges, including the difficulty of evaluating whether generated hypotheses are genuinely novel versus merely reformulations of existing knowledge. Realizing the full vision will require advances in long-horizon planning, persistent memory, self-evaluation, and the integration of diverse scientific tools into coherent agent architectures [33, 34].

8 Conclusion

This survey has charted the landscape of LLM-based difficult question synthesis, a field that has witnessed a fundamental transformation in how the research community approaches the challenge of evaluating increasingly powerful AI systems. The central finding is a paradigm shift from one-shot, prompt-driven generation toward complex, multi-stage, multi-agent creation ecosystems—a methodological evolution spanning five distinct stages (Figure 1) from primitive direct prompting through iterative self-refinement, adversarial self-play, and multi-agent verification to the autonomous synthesis ecosystems that define the current frontier. This shift is characterized by three interrelated trends.

The first trend is the move from monolithic to modular architectures. Early approaches relied on a single prompt to produce difficult questions. Contemporary pipelines—from hypothesis-driven error analysis [51] to multi-agent verification systems [54] to neuro-symbolic frameworks like Circuit-Synth [80]—decompose the task into specialized stages, each employing purpose-built techniques and tools. The second trend is the move from static to adaptive systems. MathDuels [52] demonstrates co-evolutionary difficulty through self-play; DBench-Bio [55] achieves contamination resistance through monthly regeneration from fresh literature; BenchBench [53] provides the meta-evaluation framework necessary to compare generation approaches systematically; and LiveMathematicianBench [75] introduces substitution-resistant evaluation that exposes the fragility of pattern-matching strategies. The third

trend is the move from automated to synergistic processes. ActuBench’s finding that verification is load-bearing—flagging a majority of drafted items on first pass—underscores that the human-machine interface is not merely about preference alignment but about structural quality assurance.

In specific scientific domains, mathematics synthesis has matured to incorporate formal verification and personalized problem generation [76, 77], physics and chemistry generation integrate simulation tools and retrosynthetic analysis [14], materials science leverages knowledge graphs to identify genuine research gaps [48, 49], and new domains including actuarial science [54], clinical assessment [79], time series reasoning [78], and genomics are developing specialized synthesis pipelines. The emergence of AI scientist agents signals a transition from answering questions to defining questions, a shift with profound implications for the practice of science itself.

Yet formidable challenges remain. Song et al.’s [57] empirical demonstration of widespread contamination sensitivity across multiple models reveals that the evaluation crisis is deeper than previously understood: benchmark scores may conflate exam-oriented competence with principled capability. The tension between novelty and contamination resistance demands continuous innovation in benchmark design—innovation that systems like MathDuels and DBench-Bio are beginning to deliver. The cost and scalability of expert evaluation constrain the pace of quality improvement, though multi-agent verification architectures offer a partial solution. And the ethical dimensions of automated assessment generation—equity, misuse, accountability—require sustained attention from researchers, policymakers, and the broader public.

Ultimately, the endeavor to create AI systems that can pose questions worthy of humanity’s last exam is itself the engine driving AI capability forward. LLMs are evolving from passive repositories of knowledge into active intellectual partners capable of probing the boundaries of human understanding. The intelligence arms race between question generation and question solving promises not only more capable AI but, perhaps, the discovery of genuinely new knowledge and a deeper appreciation of the limits—and the expansiveness—of human intelligence itself.

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